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Research on the prediction method of string-supported dome cable force based on digital twin technology

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Abstract

To improve the accuracy of rope force prediction and the safety of the construction process. In this paper, based on digital twin technology in the design process, digital twins of physical entities in virtual space are constructed using digital to achieve digital control and optimization of physical entities. The structural response information collected by sensors is brought into the dynamic load identification method by calculating the solution method of the dynamic load of the digital twin technology. The damping effect is considered in the load vector. The displacement is eliminated by constructing an average loss function to disperse the motion time to solve for the dynamic load imposed on the structure and obtain the load-bearing state of the structure for dome cable force prediction. To verify the feasibility of the model, the results of the cable force prediction analysis show that the overall displacement of the mesh shell displacement nodes at the bishop number 200-400 has a more uniform step distribution, and the cable force of the internal and external radial cables has a small change in the predicted structural cable force at the position of 40° angle with the long axis. Thus, it can be seen that the cable force prediction of a circular chord-supported dome with digital twin technology can break the tradition and further improve structural safety and implementability.

Keywords: Digital twin technique; Mean loss function; Load vector; Damping effect.
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1 Introduction

The string-supported dome structure is a new type of efficient large-span building structure, which is usually a new hybrid space structure system combining the upper single-layer mesh shell and the lower cable rod system [1-3]. The structure has reasonable force and cleverly combines the advantages of both single-layer mesh shell and cable dome, which is a high-performance structure with reasonable force science and is widely used in large public buildings such as stadiums and convention centers [4-7]. Given that such structures are characterized by dense crowds and large flows of people, attention needs to be paid to their structural safety, so the study of the effects of explosions, impacts and other dynamic effects of chord-supported domes cannot be missing [8-10]. The string-supported dome structure, in which the tension cables are subjected to huge tensile forces, is the core component of the structure. In terrorist attacks, construction defects and other unexpected circumstances will occur the phenomenon of broken cables; the sudden breakage of the tension cable will produce a certain dynamic impact effect on the structure, causing the vibration of the nearby rod position [11-13]. The internal force changes and the sudden impact effect will make the internal force of the rod oscillate up and down substantially, and its magnitude can far exceed the original level, so the study of the dynamic impact effect of the broken cable is very important [14-18].

Due to the continuous development and improvement of digital twin technology, the emergence and application of various large-scale and comprehensive finite element design and analysis software have effectively solved the problems of prestressed steel structures in design and analysis, construction simulation, engineering applications and other key technologies, which have led to the faster development of large-span prestressed steel structure systems. For example, intelligent energy management operation through digitalization is realized in the literature [19]. These problems are overcome by changing the microgrid configuration and operation algorithms in virtual space in various ways and performing real-time tests using digital twin technology. Propose an operational scheduling model for energy storage systems to be applied in the virtual space when building microgrids using digital twin technology. The optimal charging and discharging schedules are established and implemented using a decision tree and nonlinear autoregressive neural network model instead of existing optimization techniques. The literature [20] proposes that processing quality is the basis for ensuring product quality, reflecting the development needs and application values for achieving smart manufacturing. For the problems of low efficiency, poor timeliness and unpredictable quality control in tracing quality problems during machining, digital twin technology can provide a new intelligent solution. Firstly, a Bayesian network model for factor analysis affecting processing quality is introduced to determine the correlation and influence the weight of each factor with processing quality. Secondly, it integrates multi-source heterogeneous data to track processing quality and establishes a multi-level scalable information model and correlation mechanism.

A novel stochastic computational framework based on fractional Meyer wavelet artificial neural networks for nonlinear Singer fractional differential equations was designed in the literature [21]. The modeling advantages of wavelet artificial neural networks are used to convert differential systems into difference equations and to implement an approximation theory in the sense of mean square error to develop merit functions for nonlinear Singer fractional differential equations. The literature [22] shows that digital twin technology has been increasingly used to optimize the performance of industrial systems and is listed as one of the top ten most promising technology trends for the next decade. In it, a holistic digital twin architecture is proposed where the network is integrated along with other industrial components of the system. A digital network twin allowed closed-loop network management throughout the network lifecycle. Connections between each digital twin and its physical counterpart in the industrial system can be established. The feasibility of the proposed architecture was verified by selecting the most suitable communication mechanism to meet the real-time requirements of a flexible production system. Each model in the above literature fails to analyze its

performance by deriving performance evaluation metrics and suffers from data fragmentation, insufficient data mining and lack of fusion presentation.

Based on this, this paper applies the digital twin technique to the string-supported dome, firstly using the virtual physical, digital twin and collecting response information to find the analysis of the loaded digital twin technique. The system equations of motion are discretized by time, and the overall stiffness matrix of the multi-degree-of-freedom spatial system is used. Secondly, the average loss function is constructed to eliminate the displacement, and the dynamic load is solved by the motion time to obtain the structural load-bearing state. Finally, the sensors are arranged to collect the easy-to-measure response of the equipment in real-time and invert the load vector containing the load magnitude and action position information. The corresponding displacement column vectors can be solved by the effective stiffness matrix and the effective load vector, and the optimal inverse parameters are estimated by substituting them into the positive problem model. After combining with the digital twin technique, the comparative analysis of the mesh displacement, the X- and Y-directional support reaction forces and the radial cable force is carried out to verify the prediction results of the model, which provides data support for the prediction of the cable force of the chord-supported dome.

2 Digital twin technology

2.1 Digital twin technology infrastructure

Combined with the knowledge and understanding of the digital twin, the digital twin can be defined as a refined digital description of product entities. Digital twin technology can reflect the characteristics, behavior, formation process and performance of physical products more realistically based on the simulation experiments of digital models, can manage the data related to the whole life cycle of products, and can interact between reality and reality to realize the association and mapping of real-time collected data to the digital twin. This enables product identification, tracking and monitoring, prediction and analysis of simulated object behavior, fault diagnosis and early warning, problem location and recording through the digital twin, and optimal control, as shown in Figure 1.

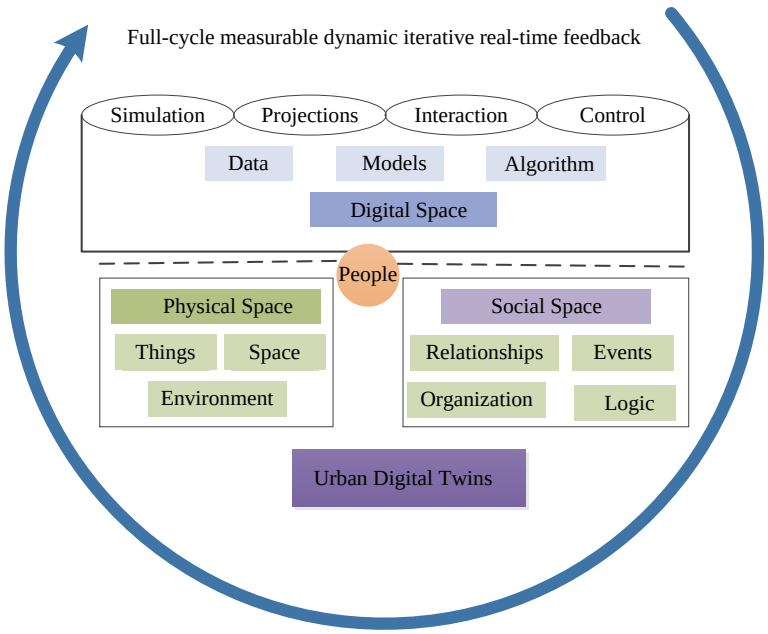


Figure 1. Digital twin technology architecture

The digital twin technology architecture is shown in Figure 1. The digital twin technology architecture can be decomposed into a professional analysis layer, virtual-real interaction layer and basic support layer according to technical characteristics, with security interconnection technology and high-performance parallel computing technology as the foundation of the digital twin, using data management technology to support data management of the whole life cycle of products, realizing refined digital representation of products through refined modeling and simulation technology, based on information physical. The data is collected in real-time based on the physical information system, and combined with data model fusion technology and interaction and collaboration technology for virtual-real interaction, thus realizing intelligent decision-making, diagnosis and prediction, visual monitoring, and optimization control. Digital twin technology is different from simulation technology. That uses software to simulate the deterministic laws and complete mechanisms of the physical world. Based on the perfect architecture of digital twin technology and its interoperability, scalability, fidelity, real-time, and closed-loop characteristics, it has been applied in many fields and achieved good results.

2.2 Load vector study

For a system with a degree of freedom N_f, considering the damping effect of the system, its equations of motion can be expressed as:

$$M\ddot{X} + C\dot{X} + KX = F(t) \tag{1}$$

In equation (1): M is the mass matrix, C is the damping matrix, K is the stiffness matrix, X is the system displacement column vector, and $F(t)$ is the system dynamic load column vector.

Mathematically, equation (1) is a second-order ordinary differential equation, which can be solved by the standard method of solving a system of ordinary differential equations. However, in finite element dynamics analysis, the order of the matrix is very high. It is generally uneconomical to use the standard method, so other effective methods are generally used in finite element dynamics analysis. The analysis is carried out in this paper based on the digital twin technique.

The system equations of motion are discretized in time with time increments of Δt and t moments as:

$$M\ddot{X}_t + C\dot{X}_t + KX_t = F_t \tag{2}$$

Assuming that the acceleration of the system at some point in the middle of moments t and $t+\Delta t$ is a constant between the acceleration values of the two moments, introduce the relationship between velocity and displacement as:

$$\dot{X}_{t+\Delta t} = \dot{X}_t + (1-\alpha)\ddot{X}_t\Delta t + \alpha\ddot{X}_{t+\Delta t}\Delta t \tag{3}$$

$$X_{t+\Delta t} = X_t + \dot{X}_t\Delta t + (\frac{1}{2}-\beta)\ddot{X}_t\Delta t^2 + \beta\ddot{X}_{t+\Delta t}\Delta t^2 \tag{4}$$

The equation of motion of the system at the moment $t+\Delta t$ is:

$$M\ddot{X}_{t+\Delta t} + C\dot{X}_{t+\Delta t} + KX_{t+\Delta t} = F_{t+\Delta t} \tag{5}$$

Formula (3), formula (4) and formula (5) can be organized to obtain:

$$\bar{K}X_{t+\Delta t} = \bar{F}_{t+\Delta t} \tag{6}$$

In equation (6): $\bar{K}$ is the effective stiffness matrix; $\bar{F}$ is the payload vector.

The corresponding displacement column vectors can be solved by the effective stiffness matrix and the effective load vector, now define the generalized load vector of the spatial system at the moment t as:

$$F(t) = \begin{bmatrix} F_1(t) \\ M_1(t) \\ \mathrm{M} \\ F_n(t) \\ M_n(t) \end{bmatrix} \tag{7}$$

$$F_n(t) = \begin{bmatrix} F_{nx}(t) \\ F_{ny}(t) \\ F_{nz}(t) \end{bmatrix} \tag{8}$$

$$M_n(t) = \begin{bmatrix} M_{nx}(t) \\ M_{ny}(t) \\ M_{nz}(t) \end{bmatrix} \tag{9}$$

In Eqs. (8) and (9): n is the system node number $F_n(t)$ and $M_n(t)$ are the system node's force and bending moment vectors n at the moment t, respectively, representing the load directions.

Accordingly, the generalized displacement vectors of the spatial system at the moment t are defined as:

$$X(t) = \begin{bmatrix} U_1(t) \\ \theta_1(t) \\ \mathrm{M} \\ U_n(t) \\ \theta_n(t) \end{bmatrix} \tag{10}$$

$$U_n(t) = \begin{bmatrix} U_{nx}(t) \\ U_{ny}(t) \\ U_{nz}(t) \end{bmatrix} \tag{11}$$

$$\theta_n(t) = \begin{bmatrix} \theta_{nx}(t) \\ \theta_{ny}(t) \\ \theta_{nz}(t) \end{bmatrix} \tag{12}$$

In Eqs. (11) and (12): $U_n(t)$ and $\theta_n(t)$ are the lateral displacement vector and the turning angle vector of the node n of the system at the moment of t, respectively.

Solving for the overall stiffness matrix K, damping matrix C and mass matrix M of the multi-degree-of-freedom spatial system, the relationship between the displacement vector and the payload vector in the spatial coordinate system can be obtained as:

$$X(t) = \bar{K}^{-1}\bar{F}(t) \tag{13}$$

In this paper, the problem of solving the inverse of the load vector, i.e., the known displacement vector at the measurement point, is solved by the inverse of the load vector containing the magnitude and location information of the load. The objective function is constructed based on the calculated displacement value $U(P,t)$ and the measured displacement value $U(t)$:

$$J(P) = \frac{1}{n}\sum_{i=1}^{n}(U(P,t)_i - U(t)_i)^2 \tag{14}$$

In formula (14): P is the calculated value of the load vector. The objective function is based on the mean square error between the calculated displacement value and the measured value at the measurement point to eliminate the effect caused by the number of elements in the displacement vector. The mean square error is averaged over the number of elements, and the objective function is then the average loss function, where n is the number of elements in the displacement vector. This treatment allows the objective function to obtain a uniform index independent of the number of elements by averaging.

In actual engineering, not all system nodes are under stress, so to optimize the solution environment and improve the solution speed, the load locus vector δ and the load value vector p are constructed. For each position information in the load, a vector is saved in vector δ, and the load value information of the nodes is saved in vector p. Therefore, during the solution process, the calculated value P of the load vector is decomposed into the load locus vector δ and the load value vector p in the solution process. This improves both the accuracy and the efficiency of the load vector solution. Denoted as:

$$P = p \, \mathrm{e} \, \delta \tag{15}$$

Using the algorithm to solve the objective function so that the objective function is satisfied, the load vector time data of the system can be obtained.

2.3 Inverse parametric construction

For loads that are difficult or impossible to measure for major projects, the load prediction accuracy can be effectively improved by using inverse problem theory and fast computational inverse methods to solve them. The covariates describing or constructing the model are screened to identify the covariates to be inverse solved. When the number of parameters is difficult to solve, sensitivity analysis can be used to reduce the number of parameters to be identified to improve the computational efficiency and accuracy of the inverse parameters. Among them, the inverse covariates can be expressed as:

$$X=(X_1^T,X_2^T,X_3^T,\mathrm{L}\ ,X_n^T)=\begin{pmatrix} x_{11} & x_{12} & \mathrm{L} & x_{1m} \\ x_{21} & x_{22} & \mathrm{L} & x_{2m} \\ \mathrm{M} & \mathrm{M} & \mathrm{M} & \mathrm{M} \\ x_{n1} & x_{n2} & \mathrm{L} & x_{nm} \end{pmatrix} \tag{16}$$

In Eq. (16), $x_{ij}\in\left[x_{\max},x_{\min}\right]$, $x_{\max}$, and $x_{\min}$ denote the upper and lower bounds of the value of x, respectively, n denotes the class of the inverse covariates, and m is the number of covariates, which is assumed to be equal for convenience.

A numerical model of the inverse parameters and the easily measurable response of the equipment is established as a positive problem solver. It can be expressed as:

$$Y_{Measurements}=G_{Positive}(X) \tag{17}$$

In Eq. (17) $G_{Positive}$ denotes the positive solver whose dimension depends on $Y_{Measurements}$ and $Y_{Measurements}$ is the positive solver prediction, which can be expressed as:

$$Y_{Measurements}=(Y_1^T,Y_2^T,Y_3^T,\mathrm{L}\ Y_k^T)=\begin{bmatrix} y_{11} & y_{12} & \mathrm{L} & y_{1l} \\ y_{21} & y_{22} & \mathrm{L} & y_{2l} \\ \mathrm{M} & \mathrm{M} & \mathrm{M} & \mathrm{M} \\ y_{k1} & y_{k2} & \mathrm{L} & y_{kl} \end{bmatrix} \tag{18}$$

In Eq. (18) k represents the total class of predicted values of the positive solver, l which is the number of predicted values in each class, and here the number of predicted values is assumed to be equal.

Deploy sensors to collect the easy-to-measure response of the equipment in real-time, compare it with the response solved by the positive problem model, and use the digital twin technique to obtain the estimation of the optimal inverse parameters. Combining Eq. (17) and Eqs. (18)~(19), the inverse problem is optimized. Estimating the model parameters using the inverse computational technique is based on the premise and condition that the positive problem model is valid. The inverse parameters can be substituted into the positive problem to effectively improve the accuracy and credibility of the model, thus laying the foundation for a more accurate equipment performance analysis.

3 Chord-supported dome cable force prediction method exploration

3.1 Design of string-supported dome structure system

The chord-supported dome structure is a new composite space structure that has become rapidly popular in the last decade. The chord-supported dome structure system belongs to the tensioned chord structure system, a branch of prestressed steel structure in space structure. The chord-supported dome mainly solves two major problems of the traditional space structure: first, it increases the overall stiffness of the structure and ensures the overall stability of the structure; second, it has the advantage of easy construction because the upper mesh shell of the chord-supported dome has a certain initial stiffness. In addition, the study shows that the chord-supported dome can reduce the support's horizontal thrust and reduce the substructure's burden as other tensioned chord structures. The chord-

supported dome is a typical non-decomposable spatial tensioned chord structure consisting of the upper mesh shell and the lower spars and cables, as shown in Figure 2.

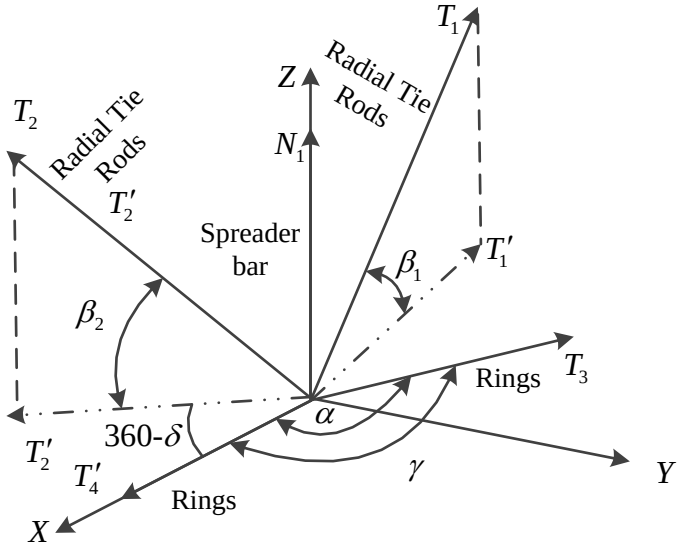


Figure 2. Schematic diagram of the structural composition of the strut dome

The structural composition of the support dome is shown in Figure 2; the upper end of each ring of props is hingedly connected to each link point corresponding to the mesh shell, and the lower end of the props is connected to the next link point of the single-layer mesh shell with radial ties, and the lower end of the props of the same ring is connected to a closed ring by the ringwise cable so that the whole structure forms a complete structural system, and the structural force transmission path is relatively clear. Under external load, the load is transferred to the lower gusset through the single-layer mesh shell at the upper end and then transferred to the cable through the gusset. After the cable is stressed, the reverse tension on the support is generated, so the thrust of the whole structure on the restrained ring beam at the lower end is greatly reduced. At the same time, due to the elastic support effect of the spar, the vertical displacement of the upper single-layer mesh shell at each link point is significantly reduced, and the static performance and stability performance of the chord-supported dome are analyzed more systematically. The influence law of parameters such as sagittal-to-span ratio, nodal stiffness of upper mesh shell, pre-stress level of ring cable, brace height and cable arrangement on structural and mechanical performance is quantitatively analyzed, and it is found that the chord-supported dome is a kind of structure with high stiffness and the characteristic of non-linearity of structure.

3.2 Numerical modeling of the chord-supported dome

During the construction of the upper mesh shell of the chord-supported dome, the support frame needs to bear the weight of the whole structure and construction load, and the deformation of the support frame during construction directly affects the installation accuracy of the mesh shell and has a relatively large impact on the prestressing construction. Therefore, the design of a string-supported dome structure construction support frame should meet the requirements of deformation control, stability and stiffness required by codes and engineering technologandso the requirements of prestressing construction. To investigate the rationality of the self-balancing turn-by-turn determination method, numerical simulations are performed for the construction process of the chord-supported dome structure, especially the prestressing application process. In this paper, based on the

digital twin technology, the numerical model of the chord-supported dome is constructed, and the material specification locations are shown in Table 1.

Table 1. Internal Force Distribution of the Cable Rod System

Location		Maximum Value	Minimum Value
Inner Circle	Ring to Cable	158.00	125.65
	Radial Steel Tie Rods	69.22	42.85
	Spreader Bar	-48.00	-41.95
Middle Circle	Ring To Cable	442.55	357.05
	Radial Steel Tie Rods	120.58	37.55
	Spreader Bar	-82.15	-62.44
Outer Ring	Ring to Cable	956.14	781.56
	Radial Steel Tie Rods	185.07	64.01
	Spreader Bar	-114.52	-72.68

The structural composition of the support dome is shown in Table I. The size, shape and material properties of the dome structural system have been determined, and the elastic modulus and flexural stiffness have been determined. According to the load vector set boundary conditions, from deformation control, stability and stiffness requirements, numerical analysis and simulation, the state variables of the dome structure system, and finally, validation and evaluation of the model, including comparison with the actual test data and model error analysis. According to the model validation results, the model is improved and extended to predict and analyze the state variables of the dome structure system. In this model, the internal force of each circle of the cable rod system increases from inside to outside, with the fastest growth of the circumferential cable force (compared with the minimum value, the middle circle is 173.8% larger than the inner circle, and the outer circle is 142.3% larger than the middle circle), and the cable force near the short axis is larger, and the brace axial pressure is smaller, while the cable force near the long axis is smaller and the brace axial pressure is larger. Under the same load combination, the vertical displacement of the chord-supported dome structure is smaller than that of the corresponding single-layer mesh shell, and the displacement produces abrupt changes at the top node of the inner ring gusset. The radial rod axial forces of the chord-supported dome structure are all smaller than those of the corresponding single-layer mesh shells, and the drop is obvious. Therefore, the radial rods of the chord-supported dome structure at the same position can be appropriately reduced in cross-sectional size compared with the single-layer shell. Compared with the corresponding single-layer shell, the axial force of the annular rod of the chord-supported dome near the center of the shell is reduced, and the axial force of the annular rod near the support area is increased. The outer ring annular rod of the chord-supported dome structure at the same position needs to increase the cross-sectional size compared with the single-layer mesh shell.

4 Chord-supported dome cable force prediction analysis

In this paper, a numerical model of a chord-supported dome based on the numerical model of a chord-supported dome is designed by combining the digital twin technology, studying the load vectors and constructing the inverse parametrization. To verify the feasibility of the model, this paper analyzes the prediction results of the cable force of the chord-supported dome numerical model based on digital twin technology by comparing the displacement of the mesh shell, the support reaction force and the cable force of the radial cable.

4.1 Comparative prediction of mesh shell displacement

Digital twin technology combines structural engineering to summarize a more intuitive structural twin architecture. First, a channel is constructed that enables real-time data transfer with physical entities. Secondly, a highly simulated digital twin modeling of the problem to be solved is performed based on real-time data. Since the absolute value of the mesh shell displacement is small, it is not normalized. The analysis calculates the data results of mesh shell displacement, support reaction force and tensile force caused by the difference in the members' specification, the prestress's size, the number of lower supports and the form of the structure. Different and different change laws, and after normalization, the differences within the resultant data can be eliminated, and the influence of the subtle differences of the basic parameters on the research results can be reduced, as shown in Figure 3.

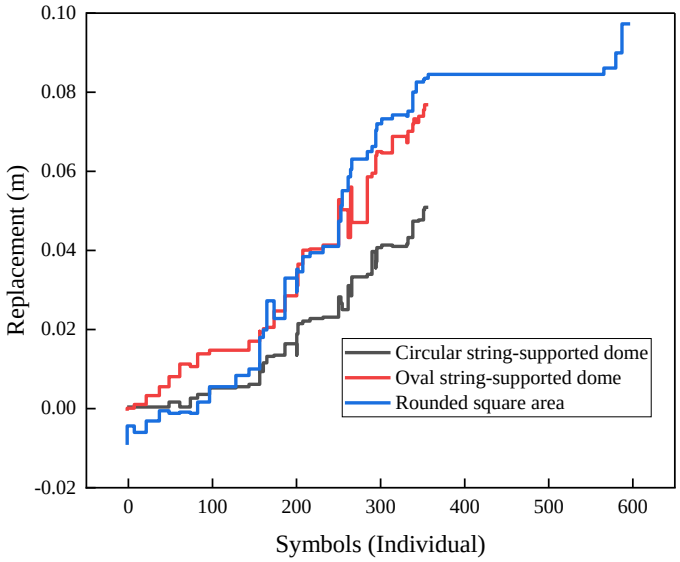


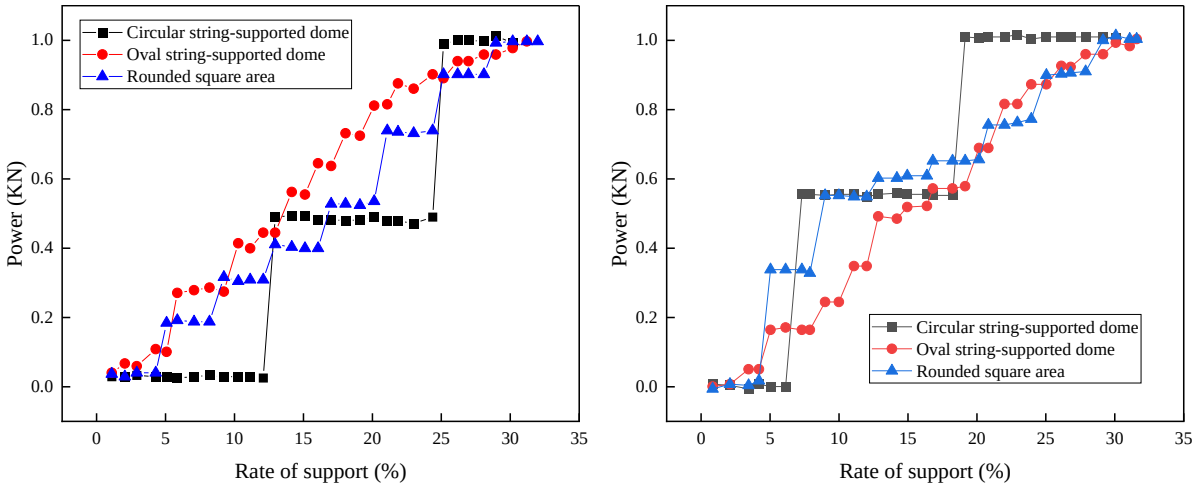
Figure 3. Comparative analysis of mesh shell displacement

The comparative analysis of the mesh shell displacement is shown in Fig. 3. When the number of bishops is 0, the displacement of the elliptical string-supported dome is -0.01, which is at the edge position, and the mesh shell node displacement of the circular string-supported dome is fixed. As the number of bishops moves from 200 to 400, the overall displacement of the casing nodes of the circular chord-supported dome shows a more uniform stepped distribution, indicating that the displacement changes uniformly along the center of the casing toward the edge. The displacement of the elliptical chord-supported dome has a non-uniform step-like variation, and the smaller and larger displacement values occupy a larger proportion, with fewer intermediate values, indicating that its displacement distribution is more non-uniform than that of the circular chord-supported dome. The mesh shell displacement of a rounded square chord-supported dome is somewhat similar to that of an elliptical one, but the degree of unevenness is larger, which indicates that its displacement distribution is more uneven. It shows that the numerical model based on the chord-supported dome is more obvious in predicting the cable force, which is a superior method and reduces the limitations and inapplicability.

4.2 Support reaction circumferential restraint

The digital twin technology has broken the independent pattern between entity, perception, modeling and application by fusing them, acquiring operational state data comprehensively with various sensing devices, and applying data fusion techniques to fuse them into the string-supported dome

structure based on the obtained accurate data. In this paper, to study the optimal loop cable prestress, the geometric method and the gusset control area method are selected to set the loop cable prestress ratio, set the X- and Y-directional reaction forces of the support constraint in the local coordinate system set the loop cable to prestress amplitude using the criterion of equal radial displacement of the support, and design two different coordinate X- and Y-directional reaction force pairs as shown in Figure 4 by cross-combining elliptical and circular chord-supported domes.



(a) X-directional support reaction force (b) Y-directional support reaction force

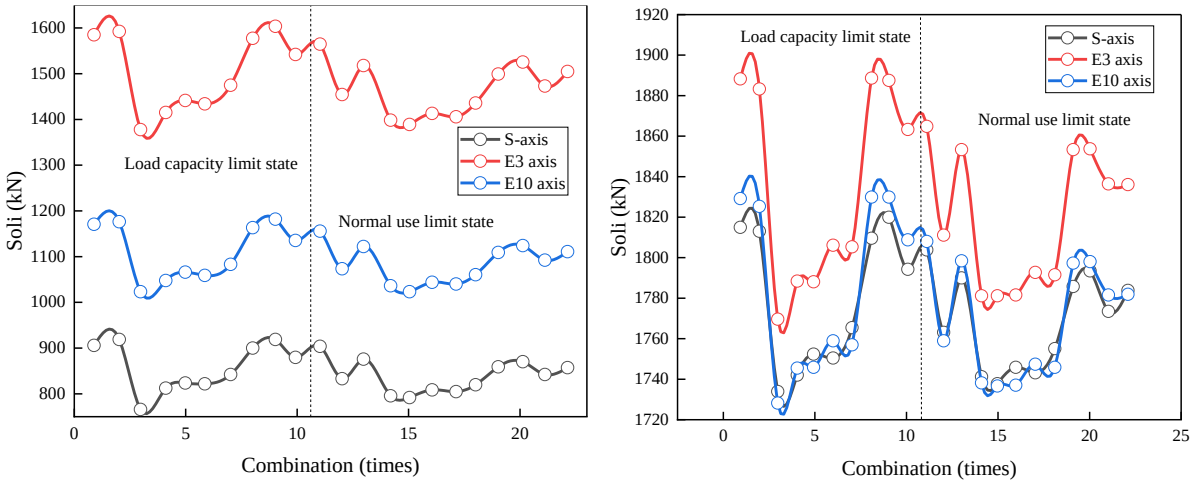
Figure 4. Comparative analysis of support reaction force

The X-directional reaction force is shown in Fig. 4(a). The X-directional reaction force of the circular chord-supported dome is distributed in three segments. However, the absolute values do not differ much (the maximum value is 30364N, and the minimum value is 26232N). The Y-directional reaction force is similar to the X-directional distribution law, but the absolute values are smaller.

The Y-directional support reaction force is shown in Figure IV(b). The X-directional reaction force (maximum value is 36665N, the minimum value is 329N) of the elliptical string-supported dome is distributed in two-point multiple steps. The step changes are more uneven, which indicates that its X-directional reaction force is symmetrically distributed along the long axis and varies more along the circumferential constraint. The Y-directional reaction force (maximum value is 41102N, the minimum value is 2394N) is similar to the X-directional distribution law similar to the X-direction, but the variation is more uneven. It can be seen that the support reaction force of an elliptical chord-supported dome varies more along the circumferential constraint and is more unevenly distributed. The X- and Y-directional reaction forces of the rounded square chord-supported dome show a four-point multi-step distribution (the maximum value is 51823N and the minimum value is 4059N in the X-direction, and the maximum value is 38609N and the minimum value is 4367N in Y-direction), indicating that the reaction forces are symmetrically distributed along the biaxial axis and vary more drastically along the circumferential constraint. The results show that the difference between the predicted and the actual force of the circular chord-supported dome and the elliptical chord-supported dome is slightly within 1-10N, indicating that the prediction of the force based on the numerical model of the chord-supported dome is less influential and the loss of prestress in the vertical reaction force of the support constraint is smaller, reflecting the insignificant change due to the shape characteristics.

4.3 Comparative analysis of rigging force of radial ropes

Digital twin simulates the behavior and characteristics of physical entities in structural engineering by building digital twins of physical entities, mapping the changes of physical entities in the real environment to achieve fault diagnosis, condition monitoring and trend prediction. To achieve these goals, digital twin technology is deeply integrated through platform-based architecture to form a closed loop of information interaction from the physical world to structural engineering. In this paper, the cable forces of radial cables are distributed by reference to an axisymmetric distribution. The cable forces of the middle radial cable in different axes do not vary much, while the difference in the cable force of the inner and outer radial cables in different axes is relatively large. The inner radial cable is in the plane of the tie beam, and the tie beam-radial cable system at each axis is relatively independent, so the tensile force of the inner radial cable is close to the tensile force. While the outer radial cable is connected with the ring cable, with an obvious group cable effect, each radial cable will affect the other. The difference in cable force at different positions is large, so the inner and outer radial cable force changes are shown in Figure 5.



(a) Variation of inner radial cable force (b) Variation of outer radial cable force

Figure 5. Variation of cable force

The variation of the inner radial cable force, as shown in Figure V(a), can be seen that the variation range of the maximum downward vertical displacement of the structure under the limit state of normal use is -79.61 mm to -257.09 mm, with a variation of about 180 mm, and the cable force of the ring cable and the stress of the structure drop significantly under the wind load. The observation of combinations 4 and 5 and combinations 6 and 7 also shows that the structure's response to the warming and cooling of 25 degrees Celsius is not very obvious. It also shows that the change in the ring cable force under normal use is insignificant.

The change of the outer radial cable force is shown in Figure V(b), which shows that the radial cable force is distributed axisymmetrically. In general, the outer radial cable force is relatively uniform. In contrast, the inner radial cable force is maximum at the E3 and E17 axes, which is about the position of 40° angle with the long axis. The inner and outer radial cable forces at the S axis (long axis), E3 axis (40° angle axis) and E1 axis (short axis) were extracted to understand the distribution of radial cable forces in different combinations in more detail. The results show that the prediction of the ring cable force is not much different from the actual one, and the variation is between 10mm, which indicates that the prediction of the structural cable force under different combinations in the numerical

model based on the chord-supported dome is not very big, so the structure has good flexibility and toughness, which can improve the internal structural force and provide safety reserve.

5 Conclusion

This paper combines the digital twin technique with the chord-supported dome to solve for dynamic loads to obtain the structural load carrying the state. The response information is collected using the virtual physical and digital twin, and the average loss function is constructed to predict the cable force of the chord-supported dome. In the prediction process, a numerical model of the chord-supported dome based on the digital twin technique is designed and analyzed to verify the model's practicality. The following conclusions are obtained:

1) When the mesh shell displacement elephant number movement is not obvious to be an uneven step change, the smaller displacement value and larger displacement value in the circular chord-supported dome occupy a larger proportion. The middle value is less, indicating that its displacement distribution is more uneven than that of the circular chord-supported dome, indicating that the prediction of the cable force is more obvious in the numerical model based on the chord-supported dome. The method is superior to reducing the limitation and inapplicability.

2) When the maximum values of the X-direction and Y-direction are 51823N and 38609N, the reaction force is symmetrically distributed along the biaxial axis, and the change along the circumferential constraint is more drastic. It indicates that the influence of cable force prediction in the numerical model based on a chord-supported dome is smaller, and the loss of prestress in the vertical reaction force of the support constraint is smaller.

3) In the internal and external radial cable force at the S axis (long axis), E3 axis (40° pinch axis) and E1 axis (short axis), the structural cable force prediction changes less, indicating that the structural cable force prediction under different combinations in the numerical model based on the chord-supported dome is not very variable, so the cable force prediction of the chord-supported dome in this paper has better accuracy.

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